Mamadou THIAO

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PhD • Senior Software Engineer

PhD-level mathematical and software engineer with extensive experience building and reviewing complex software, algorithms, mathematical models, and data-processing systems across medical imaging, industrial tools, autonomous vehicles, dental health, geoscience, finance, and scientific computing. Work through advanced reasoning, careful technical judgment, precise problem solving, clear communication with experts, and hands-on experience with C++ and Python development.

Selected publications & talks

efficient approaches for ℓ2-ℓ0 regularization at **Springer Applied Intelligence** • sparse principal component analysis at **ICML** • l0 norm programs at MCO • sparse approximation at AFG • sparse Fisher discriminant analysis at EURO • feature selection at EURO • sparse logistic regression at ROADEF • workshops on optimization and machine learning • technical day on autonomous vehicles and mobile ground robotics • technical day on polyhedra and combinatorial optimization.

Skills, Tools & Experience

C++ • Python • Software Development • Software Architecture • Algorithms • Applied Mathematics • Machine Learning • Mathematical Programming • Computational Geometry • 3D Data • Point Clouds • LiDAR Data • Image Processing • DICOM • Computer Vision • Photogrammetry • Surface Reconstruction • Signal Processing • Data Mining • Sparse Modeling • Optimization • Statistical Modeling • Database Systems • Networked Software • Visualization • LabVIEW • MATLAB • Technical Communication • Conference Presentations • Cross-Functional Expert Collaboration • Engineering Mentorship

Experience

Indilo - Software Engineer | Thiès, Sénégal | may 2025 – july 2026
- Prototyped scientific software from initial technical concepts through architecture, design, and implementation.
- Designed and developed software solutions for numerical and scientific computing problems.
- Applied mathematical reasoning and software engineering judgment to turn complex technical requirements into working prototypes.

Solent - Software Engineer | Nantes, France | june 2022 – march 2025
- Developed software for medical imaging and intelligent industrial tightening tools.
- Worked with 3D data, images, and DICOM data in software development and technical problem solving.
- Applied careful engineering judgment to complex software requirements in medical and industrial environments.

Navya Group - R&D Engineer | La Défense, France | january 2019 – october 2021
- Developed mapping software using 3D data, point clouds, NURBS, storage systems, and LiDAR data.
- Advanced to Software Architect for the perception division, supporting mapping, localization, and obstacle teams.
- Coordinated technical direction across multiple engineering teams and communicated complex software decisions clearly.

Philog - Software Engineer | Ile-de-France, France | may 2014 – december 2018
- Developed face reconstruction and dental restoration software using 3D, photogrammetry, camera data, and DICOM.
- Developed geological modeling and subsurface layer decompaction software using elasticity and seismic data.
- Applied algorithms, mathematical modeling, and software engineering across distinct technical domains.

OptoPartner - Software Engineer | Ile-de-France, France | july 2012 – january 2014
- Converted complete software pipelines from LabVIEW and MATLAB to C++.
- Developed software involving LiDAR data, signal processing, instrumentation, trajectory planning, databases, networks, visualization, and command systems.
- Integrated multiple technical components into production-oriented software pipelines.

LITA - PostDoctoral Researcher | Metz, France | december 2011 – june 2012
- Developed Markowitz-based portfolio selection software using sparse modeling and investment-fund data.
- Applied mathematical optimization and data-driven reasoning to financial modeling problems.

INSA - PhD Researcher | Rouen, France | 2011
- Conducted doctoral research in programming, algorithms, sparse modeling, and data mining.
- Published research and presented technical work at conferences and professional events.

CORIA - Engineering Internship | Rouen, France | 2007
- Extended a shock-tube gas-flow simulation using Navier-Stokes, BGK-Burnett, and WENO methods.

INSA - Capstone Project | Rouen, France | 2007
- Constructed balanced and highly non-linear Boolean functions for a cryptography project.

UGB – 4-Year Bachelor Thesis | St-Louis, Sénégal | 2005
- Developed robust estimators using influence-function methods.

Education

PhD, Applied Mathematics | INSA, Rouen, France | 2011
MSc, Mathematical and Software Engineering | INSA, Rouen, France | 2007
MSc, Fundamental and Applied Mathematics | INSA, Rouen, France | 2007
4-Year BSc, Applied Mathematics and Computer Science | UGB, St-Louis, Sénégal | 2005

